Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

The dataset has **167,826 rows** and **9 columns**, with the following key characteristics:

Key Columns:

- **Country/Region**: Name of the country or region.
- **UNICEF_Region**: UNICEF's regional classification (some missing values here).
- **Indicator**: Describes the HIV-related metric (e.g., prevalence, number of cases).
- **Year**: Numeric year (stored as float, likely should be integer).
- **Sex**: Male, Female, or Both.
- **Age**: Age group (e.g., 0–4, 10–19, etc.).
- **Value**: Main quantitative figure (e.g., count or rate); currently a string and should be numeric.
- **Lower** and **Upper**: Confidence interval bounds; also stored as strings and should be numeric.

Missing Data:

- `UNICEF_Region` is missing in **10,432 rows**.
- `Lower` and `Upper` bounds are missing in **57 ro**

I'm ready to proceed with analysis—just give me a specific analytical question you'd like answered from this dataset. [>-]

Develop two models to simulate reduced and increased variability in projected HIV incidence among children and adolescents. Specify key indicators and the statistical or epidemiological techniques used to quantify variability

Summary of the Models:

We developed two simulated models to explore variability in projected **HIV incidence among children and adolescents** using historical variability as a baseline:

✅ **Key Indicator:**

- **HIV incidence** (filtered using keywords in the indicator name).
- Age and sex-specific values.

📊 Modeling Techniques:

1. **Baseline Variability**:
   - Calculated **standard deviation (Std_Baseline)** of incidence values per country, age group, and sex from historical data.
2. **Reduced Variability Model**:
   - Simulated by drawing values from a **normal distribution** centered on the historical mean, but with **50% of the baseline standard deviation**.
3. **Increased Variability Model**:
   - Simulated similarly, but with **150% of the baseline standard deviation**.

You can now explore the table showing:

- Baseline standard deviation
- Mean historical incidence
- Simulated incidence values under both reduced and increased variability models

Let me know if you'd like visualizations or to explore specific countries, regions, or trends. [>-]

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